

Markov Chains for Weather Prediction

Pedro Matheus Figueredo Cristaldo

Mateu Magem Ribas

Marta Viñas Lóbez

Probabilistic Methods

Master's degree in Machine Learning and Cybersecurity for Internet Connected Systems

May 29, 2025

Abstract

This project applies a first-order Markov Chain model to short-term weather prediction using historical categorical data from the Seattle Weather Prediction dataset. By modeling transitions between four weather states—**rain**, **sun**, **snow**, and **fog**, we constructed an empirical transition matrix and evaluated its predictive performance through stochastic simulations. The simulated matrices were compared to the empirical one using error heatmaps and the Frobenius norm, revealing consistent convergence as the number of iterations increased. The results confirm that a simple Markov model can effectively replicate short-term weather dynamics, offering an interpretable and computationally efficient alternative to complex forecasting systems. We discuss its strengths, limitations, and potential extensions for improved realism and accuracy.

1 Introduction

Accurate weather prediction plays a critical role in diverse sectors such as agriculture, transportation, disaster management, and daily human activities. Traditional forecasting techniques often rely on numerical weather prediction (NWP) models, which simulate physical processes in the atmosphere using complex mathematical equations. Although these models can be precise, they are computationally intensive and require vast amounts of real-time meteorological data, including wind velocity, humidity, pressure, and temperature.

As an alternative to physics-based approaches, statistical models offer simpler, more interpretable methods for forecasting, especially when computational resources or high-resolution data are limited. Among these, Markov Chains present a powerful probabilistic framework for modelling sequences of dependent events. Their strength lies in capturing temporal dependencies using transition probabilities between discrete states. In the context of weather forecasting, this means using historical weather conditions to estimate the likelihood of future states such as rain, sunshine, or cloud cover.

In this project, we explore the application of discrete-

time Markov Chains for short-term weather prediction. Specifically, we aim to develop a model that uses historical daily weather data to learn the probabilistic transitions between defined weather states. This enables us to forecast the most probable weather condition for the next day, based solely on the current observation. The model assumes the Markov property, that the future state depends only on the present state, which simplifies the prediction task while still reflecting realistic short-term patterns.

Our approach involves collecting and preprocessing historical weather data, discretising it into a finite set of observable states, and constructing a transition probability matrix. We then use this matrix to simulate weather sequences and evaluate the predictive performance of the model. The project also examines the steady-state behaviour of the Markov Chain, offering insights into long-term weather trends.

The remainder of the report is structured as follows. In the *Related Work* section, we review prior research on the use of Markov models in weather forecasting. The *Problem Description* section formally defines our modelling framework. The *Methodology* details the data preparation and implementation steps, followed by the *Results and Analysis*, where we present and interpret our findings. Finally, we discuss the implications and limitations of our approach before concluding with suggestions for future improvements.

2 Related Work

The use of Markov Chains in weather prediction has been a topic of interest for decades, particularly due to their ability to model systems that transition from one state to another probabilistically. In the context of meteorology, where weather conditions evolve dynamically over time and often exhibit temporal dependencies, Markov Chain models offer a natural framework to simulate and predict future states based on past observations.

One of the earlier and foundational applications of Markov Chains to meteorology can be traced to the work by Gabriel and Neumann [1], who used a first-order Markov process to model daily rainfall. In their study, they demonstrated that the probability of rainfall on a

given day was significantly dependent on the weather conditions of the previous day. Their model successfully captured the binary wet/dry state transitions in a stochastic framework and established that such simple models could outperform static probabilistic baselines. Though limited in predictive power for long-term forecasts, the model proved effective for short-term binary predictions and inspired further research into discrete-state weather modelling.

In more recent years, studies have extended the basic Markov Chain model to multi-state systems, where states represent a richer set of weather conditions (e.g., sunny, cloudy, rainy, stormy). For instance, Koushik and Deekshatulu [2] proposed a Markov Chain-based model for multi-state weather prediction, incorporating regional climate characteristics and transition matrices derived from multi-year historical data. Their approach enabled them to estimate not only the next day’s weather condition but also seasonal probabilities of certain weather events. The study revealed that incorporating higher resolution in state definitions improves predictive accuracy, although it also introduces complexity in computing reliable transition probabilities, particularly when data is sparse.

Another noteworthy contribution is the work by Balaji and Nagarajan [3], which explored the application of higher-order Markov Chains in modelling temperature and rainfall sequences. They argued that while first-order models are easier to implement and interpret, they may not capture the persistence patterns commonly observed in real weather data, such as prolonged periods of rainfall or drought. By using second- and third-order chains, the authors were able to achieve better fits to observed sequences, highlighting the trade-off between model simplicity and representational power. They also evaluated model performance across different climatic zones in India, noting that regions with higher variability benefited more from higher-order models.

Collectively, these studies demonstrate the versatility and effectiveness of Markov Chain models in capturing short-term weather dependencies. They also point to the importance of careful model design, including the choice of state definitions, order of the Markov process, and the quantity and quality of historical data, in determining predictive success. While these models are generally not used for long-range forecasting, they remain a useful tool in applications such as agricultural planning, water resource management, and educational demonstrations of probabilistic forecasting.

3 Problem Description

Weather prediction is inherently a temporal and probabilistic process, often influenced by complex atmospheric interactions. Traditional numerical methods, while accurate, are computationally expensive and require detailed

physical data. As a simpler alternative, probabilistic models such as Markov Chains can be employed to capture the statistical patterns in historical weather data. This project aims to develop a discrete-time Markov Chain model to predict daily weather patterns based on prior weather conditions. The motivation is to explore how a relatively simple mathematical model can be used to simulate and forecast real-world stochastic processes such as weather changes.

3.1 Markov Chains and the Weather Forecasting Problem

A Markov Chain is a type of stochastic process characterised by the “memoryless” property, which refers to the probability of transitioning to the next state depends only on the current state and not on the sequence of states that preceded it. Mathematically, a first-order Markov Chain is defined as follows:

$$P(X_{n+1} = j \mid X_n = i, X_{n-1} = i_{n-1}, \dots, X_0 = i_0) \quad (1)$$

$$= P(X_{n+1} = j \mid X_n = i) \quad (2)$$

We aim to model daily weather using a discrete-time, finite-state Markov Chain. This stochastic model describes a sequence of random variables $X_0, X_1, X_2, \dots$, where each X_n denotes the weather state on day n . That is, the future state depends only on the present, not the entire history.

3.2 State Space and Transition Matrix

We define a finite state space $S = \{s_1, s_2, \dots, s_m\}$, where each s_i represents a distinct weather condition such as **Sunny**, **Rainy**, or **Cloudy**. From historical data, we estimate a transition probability matrix $P \in R^{m \times m}$, where the entry p_{ij} denotes the probability of transitioning from state s_i to state s_j in one day:

$$P = \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1m} \\ p_{21} & p_{22} & \cdots & p_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ p_{m1} & p_{m2} & \cdots & p_{mm} \end{bmatrix}, \quad \sum_{j=1}^m p_{ij} = 1, \quad \forall i \quad (3)$$

Each row of P corresponds to a probability distribution over the next possible weather states, given the current condition.

3.3 Prediction, Simulation, and Steady-State Behaviour

The primary use of the model is to perform one-step-ahead predictions: given today’s weather, we infer the most likely state for tomorrow. Additionally, by simulating sequences over multiple time steps, we can generate realistic synthetic weather patterns and examine long-term tendencies.

We also analyse the steady-state behaviour of the chain, solving for the stationary distribution π , which satisfies:

$$\pi P = \pi, \quad \text{with} \quad \sum_{i=1}^m \pi_i = 1 \quad (4)$$

This distribution reveals the long-term proportion of time the system is expected to spend in each state, independent of the starting point.

3.4 Scope and Assumptions

In this project, we formulate the weather prediction problem as a classification task in which, given the current weather condition, we estimate the most probable weather condition for the following day. Historical weather data is used to estimate the transition probabilities between states, and the resulting transition matrix forms the basis of our prediction model.

We assume that weather patterns exhibit short-term dependencies that can be effectively modelled using a first-order Markov Chain. While this approach does not capture long-term correlations or external climatic influences, such as pressure systems or humidity levels, it provides a useful and computationally efficient approximation of real-world weather dynamics.

This problem formulation enables us to simulate weather sequences, calculate steady-state distributions, and make short-term predictions using historical data alone. The ultimate goal is to evaluate the predictive capabilities of this Markov model and discuss its applicability, strengths, and limitations in the context of weather forecasting.

4 Methodology

4.1 Data Collection and Preprocessing

We used a publicly available dataset from Kaggle titled *Seattle Weather Prediction Dataset*, which provides daily weather records over several years. Each record includes a date and categorical weather label, along with additional meteorological features like precipitation, temperature, and wind speed.

Since our model focuses solely on the sequence of weather conditions, we dropped the numerical weather features and retained only the date and the **weather** column. Additionally, we performed a simplification of labels by treating **drizzle** as **rain** to reduce sparsity and improve state consistency. This preprocessing step led us to work with four discrete weather states:

rain, sun, snow, and fog

We replaced the original date values with simple integer indices to facilitate state transition computations.

4.2 Definition of States and Sequence Construction

Each weather label in the dataset was mapped to one of the four predefined states. A sequence of states was then created by iterating over the dataset in chronological order, pairing each day’s label with the label of the following day. This yielded a list of consecutive state transitions such as:

(rain, sun), (sun, sun), (sun, rain), ...

These pairs form the empirical basis for estimating transition probabilities.

4.3 Transition Matrix Estimation

We constructed a first-order Markov Chain where the transition probability matrix P describes the likelihood of transitioning from one weather state to another. For each pair (i, j) , we counted how many times state i was followed by state j , then normalised the row totals to obtain conditional probabilities:

$$p_{ij} = \frac{\text{Number of transitions from state } i \text{ to } j}{\text{Total transitions from state } i}$$

This process was implemented both in a manual (explicit counters) and automated (dictionary-based) manner for validation. The resulting matrix $P \in \mathbb{R}^{4 \times 4}$ represents the transition probabilities among the four defined weather conditions.

To ensure completeness, we added zero-probability entries for rare or missing transitions (e.g., **fog** to **snow**) to maintain a consistent matrix structure. Each row was verified to sum to 1.

4.4 Weather Simulation with Markov Chains

Once the transition matrix was established, we simulated weather sequences using stochastic sampling. Starting from an initial state (e.g., **rain**), we used the corresponding row in the matrix as a probability distribution to randomly select the next state. This process was repeated over a fixed number of iterations (e.g., 100 steps) to generate synthetic sequences.

For each simulation:

- We recorded the generated sequence of weather conditions
- Recalculated the transition frequencies from the synthetic sequence
- Compared the simulated transition probabilities to the empirical ones

This allowed us to assess whether the Markov model reproduces statistically consistent behaviour.

4.5 Validation and Probabilistic Analysis

The final step in the methodology involved validating the model by:

- Counting the number of each state in the simulated chain
- Computing transition frequencies from the simulated sequence
- Normalising and comparing these to the real-world probabilities from the original dataset

We confirmed that both empirical and simulated transition matrices adhered to the stochastic requirement, rows summing approximately to 1, and that the Markov Chain captured key statistical patterns present in the historical data. These observations support the viability of Markov Chains for short-term weather simulation and probabilistic forecasting.

5 Results and Analysis

5.1 Empirical Transition Matrix

Using the cleaned and processed weather dataset from Seattle, we constructed an empirical transition matrix T representing the observed probabilities of transitioning from one weather state to another. The four states considered were: **rain**, **sun**, **snow**, and **fog**. The matrix is row-stochastic, with each row summing to 1, and is visualized below:

$$T = \begin{bmatrix} 0.69884726 & 0.22910663 & 0.01585014 & 0.05619597 \\ 0.25665102 & 0.68231612 & 0.00782473 & 0.05320814 \\ 0.42307692 & 0.19230769 & 0.38461538 & 0. \\ 0.32673267 & 0.3960396 & 0. & 0.27722772 \end{bmatrix}$$

To make the transition structure more interpretable, we visualized the matrix as a directed graph with labeled edges indicating rounded transition probabilities. This graph clearly reflects key tendencies, such as the high self-transition probability of sunny and rainy days, and frequent transitions from snow or fog to rain.

5.2 Simulated Matrices and Convergence

To assess the fidelity and robustness of the Markov model, we performed multiple stochastic simulations of weather sequences, starting from an initial condition and applying transition probabilities iteratively. We generated transition matrices T_n for various numbers of iterations $n \in \{25, 50, 100, \dots, 100000\}$. Each T_n was computed by tracking the state transitions observed in the simulated sequence and normalizing to form a probability matrix.

As expected, the simulated matrices initially deviate from the empirical matrix due to stochastic noise. However, as the number of iterations increases, the simulated

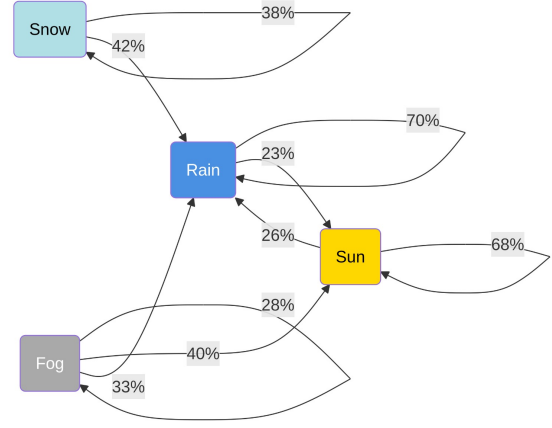


Figure 1: Graphical representation of the empirical transition probabilities.

matrices converge toward the original T . This behavior is visualized through a sequence of error heatmaps showing the absolute element-wise differences between T and each T_n :

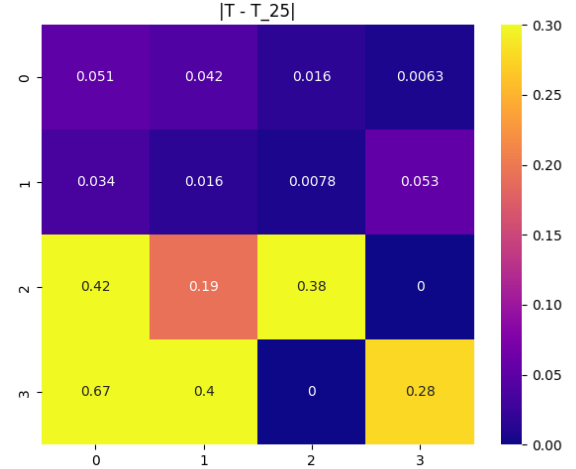


Figure 2: Error heatmap for 25 iterations

The heatmaps reveal a clear pattern: with larger simulation sizes, the element-wise error diminishes, especially in the frequently observed transitions (e.g., rain $\rightarrow$ rain, sun $\rightarrow$ sun). Rare transitions (e.g., snow $\rightarrow$ fog) converge more slowly due to low frequency in both real and simulated data.

5.3 Frobenius Norm Analysis

To quantify the convergence of T_n to T , we computed the Frobenius norm of the difference between each simulated matrix and the empirical matrix. This norm aggregates the squared differences across all entries and provides a single scalar error metric:

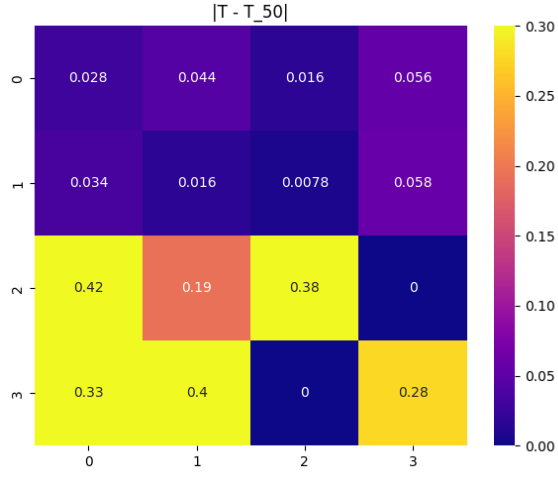


Figure 3: Error heatmap for 50 iterations

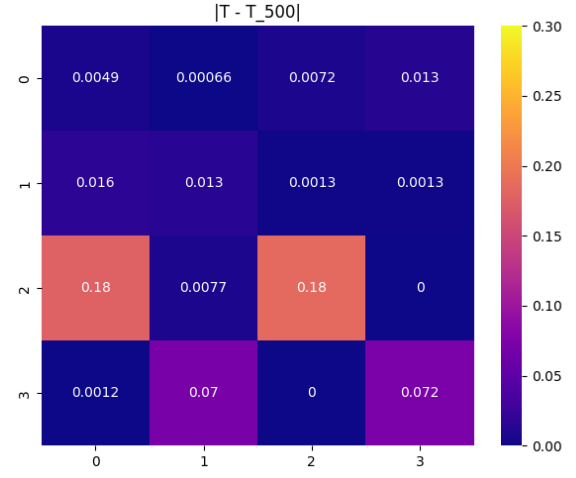


Figure 6: Error heatmap for 500 iterations

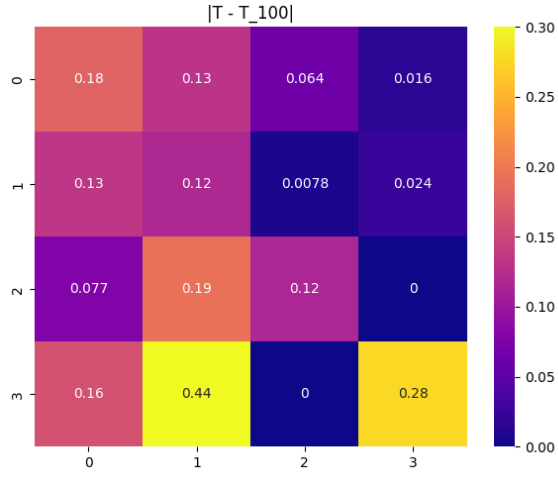


Figure 4: Error heatmap for 100 iterations

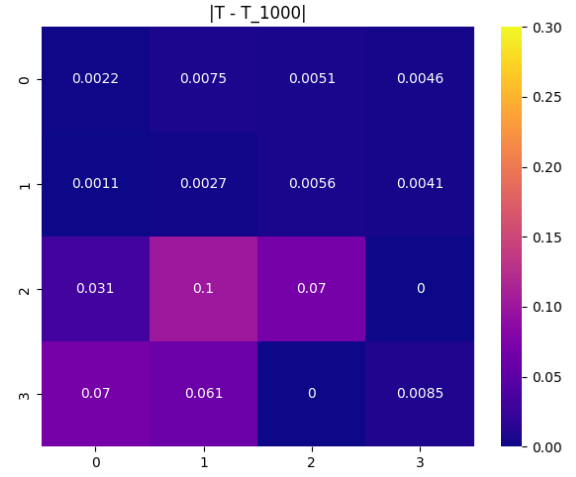


Figure 7: Error heatmap for 1000 iterations

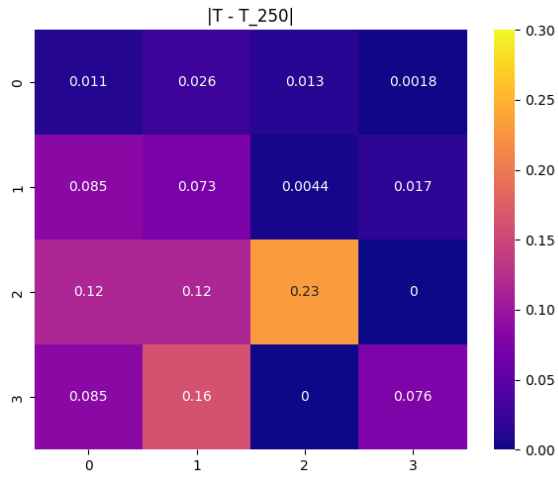


Figure 5: Error heatmap for 250 iterations

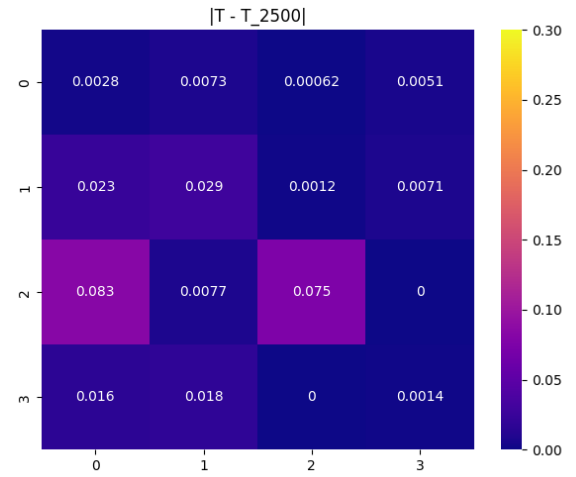


Figure 8: Error heatmap for 2500 iterations

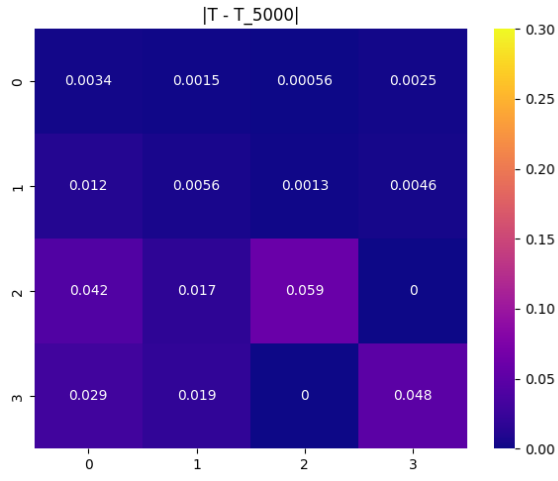


Figure 9: Error heatmap for 5000 iterations

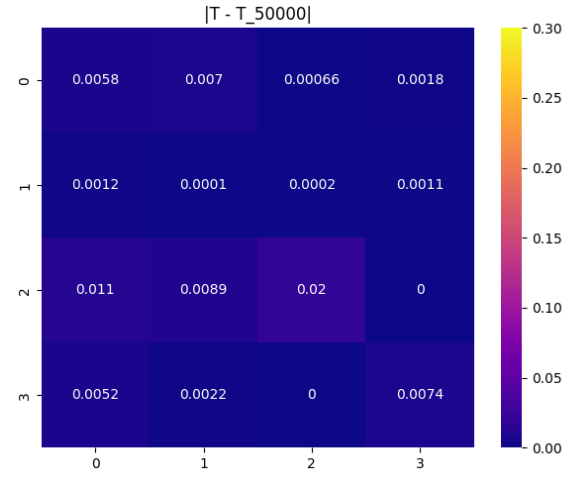


Figure 12: Error heatmap for 50000 iterations

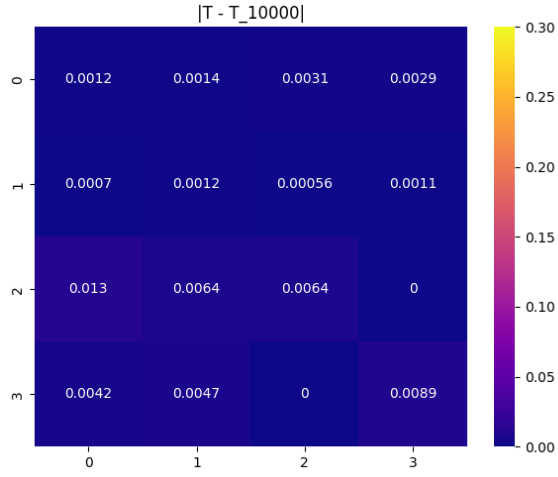


Figure 10: Error heatmap for 10000 iterations

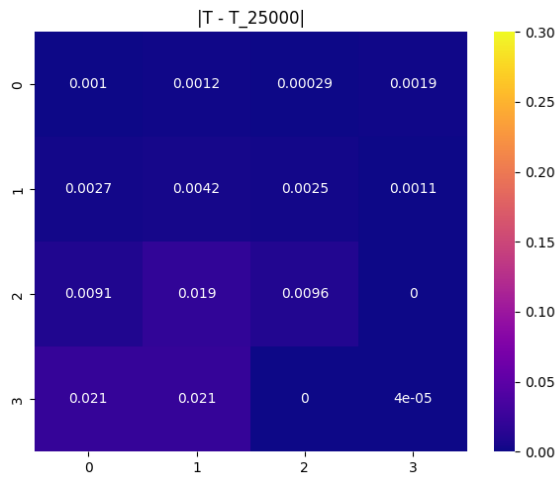


Figure 11: Error heatmap for 25000 iterations

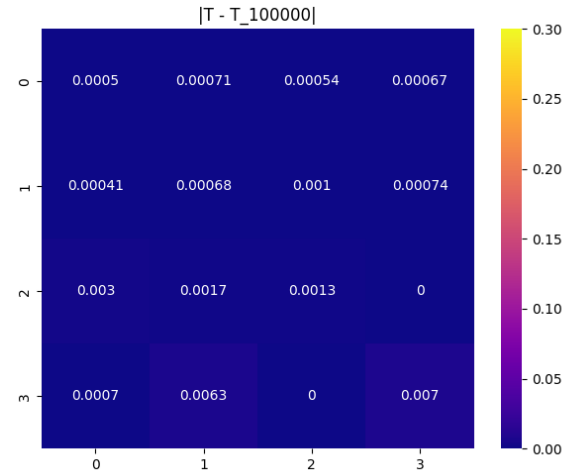


Figure 13: Error heatmap for 100000 iterations

$$\|T - T_n\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n (T_{ij} - T_{n,ij})^2}$$

We plotted the Frobenius norm as a function of the number of iterations n , using a logarithmic x-axis to highlight convergence trends across scales:

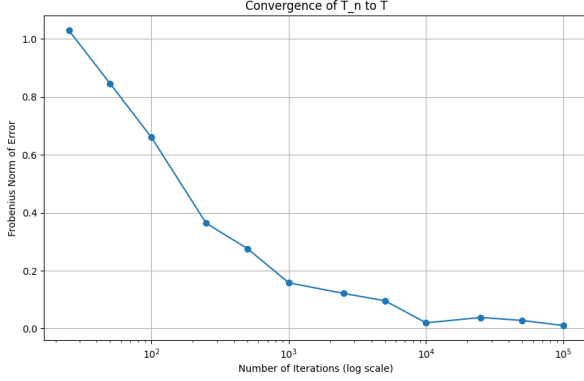


Figure 14: Frobenius Convergence from T_n to T

The results demonstrate a clear convergence trend: as n increases, the Frobenius norm decreases monotonically, approaching zero. This supports the theoretical expectation that longer Markov simulations will approximate the true transition probabilities more closely.

5.4 Interpretation and Insights

Our analysis confirms that:

- The empirical transition matrix accurately captures real-world short-term dependencies between weather states.
- The Markov Chain simulation faithfully replicates these dependencies over time, especially as the number of iterations increases.
- The convergence behavior is strongest in high-frequency transitions and slower in underrepresented transitions, which is expected given their lower empirical support.

These findings validate the use of first-order Markov Chains for short-term weather simulation and reinforce the importance of sufficient sample size in Monte Carlo-style simulations for probabilistic modeling.

6 Discussion

The results of our study demonstrate the potential of first-order Markov Chains for modeling and simulating short-term weather patterns using categorical historical data. The empirical transition matrix derived from the Seattle

weather dataset clearly captures dominant weather dependencies, such as the strong likelihood of sunny days following other sunny days and the persistence of rainy weather in the Pacific Northwest.

One of the key insights from our simulations is the rapid convergence of the synthetic transition matrices T_n to the empirical matrix T . The visual inspection through error heatmaps and the quantitative Frobenius norm analysis consistently show that as the number of simulation iterations increases, the generated matrices closely approximate the real-world transition structure.

However, our approach also highlights certain limitations. The convergence is less stable in transitions involving rare states, such as snow or fog, due to their lower occurrence in the original dataset. This sparsity introduces higher variance in simulated transition probabilities involving these states. Moreover, the first-order Markov assumption simplifies the model by ignoring dependencies beyond the immediate previous day. In practice, weather patterns often exhibit longer-term dependencies (e.g., multi-day storms or seasonal persistence) that are not fully captured by our model.

Another consideration is the model’s assumption of stationarity, that transition probabilities remain constant over time. While this works reasonably well in our dataset, real-world weather systems often evolve due to climate changes, long-term trends, or anthropogenic factors. Thus, the fixed transition matrix may be less suitable for long-term forecasting or under rapidly changing climate conditions.

Despite these simplifications, the model provides an interpretable, data-driven framework that effectively reproduces short-term weather behavior and offers insights into system dynamics. Its low computational requirements and transparency make it well-suited for educational use, rapid prototyping, and settings where only categorical data is available.

7 Conclusion and Future Work

In this project, we developed a discrete-time, first-order Markov Chain model for predicting weather patterns based on historical categorical data. Using the Seattle weather dataset, we constructed an empirical transition matrix, simulated synthetic weather sequences, and evaluated the convergence of the simulated transitions back to the empirical distribution.

Our results demonstrate that even a simple probabilistic model can capture meaningful dependencies in daily weather conditions. We confirmed the reliability of the Markov model through both visual (heatmaps) and quantitative (Frobenius norm) analyses, validating its ability to replicate short-term dynamics with reasonable accuracy.

For future work, several extensions are possible:

- **Higher-order Markov Chains:** Incorporating

memory of more than one previous state could better capture persistence in weather sequences and improve prediction for rarer events.

- **Incorporating seasonal effects:** Transition probabilities could be computed separately for different months or seasons to capture time-dependent dynamics and improve generalization.
- **Hybrid models:** Combining Markov Chains with machine learning methods (e.g., HMMs or RNNs) could allow for better handling of sequential dependencies and incomplete data.
- **Application to other locations:** Repeating the experiment using weather data from other geographical regions could test the generality and adaptability of the model.

Ultimately, our work confirms that Markov Chain models can serve as an effective tool for weather pattern analysis, especially when interpretability, simplicity, and speed are important design considerations.

Acknowledgments

We thank our course instructor for his patient teaching. Work distribution:

- Pedro Matheus F.: Implement the experiment.
- Mateu M.: Wrote analysis scripts.
- Marta V.: Wrote the documentation.

References

- [1] K. R. Gabriel and J. Neumann, “A markov chain model for daily rainfall occurrence at tel aviv,” *Quarterly Journal of the Royal Meteorological Society*, vol. 88, no. 375, pp. 90–95, 1962.
- [2] P. Koushik and B. L. Deekshatulu, “Markov chain models in daily weather forecasting,” in *Proceedings of the International Conference on Computer Science and Information Technology*, pp. 35–39, IEEE, 2007.
- [3] R. Balaji and R. Nagarajan, “Weather prediction using higher order markov chain models,” *International Journal of Computer Applications*, vol. 1, no. 19, pp. 1–4, 2010.

A Implementation Details

A.1 Code Structure

The code for this project is organized as a Jupyter Notebook with the following sections:

1. **Data Loading:** Uses the `kagglehub` package to download the Seattle weather dataset.
2. **Preprocessing:** Cleans the dataset, simplifies labels (e.g., treating drizzle as rain), and retains only relevant columns (`date` and `weather`).
3. **Transition Matrix Construction:** Creates sequences of consecutive weather states and calculates the empirical transition matrix using both manual counting and a dictionary-based approach.
4. **Simulation:** Implements a weather simulator using the transition matrix and generates synthetic sequences for different iteration sizes.
5. **Evaluation:** Computes simulated transition matrices, plots error heatmaps, and calculates the Frobenius norm for convergence assessment.

The code is written in Python and uses standard libraries including `pandas`, `numpy`, `matplotlib`, `seaborn`, and `random`. All key computations and visualizations are fully documented and reproducible within the notebook.

A.2 Code Availability

The full source code, including the notebook, generated plots, and simulations, is delivered together with this document. It is organized to enable full reproduction of all results and figures presented in this report.